10/7/24

§ 3.3 (continued)

(Solution to LSP)

$$Ax = b$$

A: $n \times m$, $n \gg m$

b: in $\mathbb{R}^n$

The task is to solve this system even if does not have a solution (inconsistent). We'll do this in the sense of finding

$$x \in \mathbb{R}^m$$

such that

$$\|r\|_2 = \|\underbrace{Ax - b}_{r}\|_2 \text{ is minimal}$$

Achieved via QR.

<u>Theorem</u> 3.3.3 Let $A \in M_{m,n}(\mathbb{R})$, $n > m$. Then there exist unique orthogonal $Q \in M_n(\mathbb{R})$ and $R = \begin{bmatrix} \hat{R} \\ 0 \end{bmatrix} \in M_{n,m}(\mathbb{R})$, where $\hat{R} \in M_m(\mathbb{R})$ is upper triangular such that $A = QR$.

<u>Proof</u> Define $\breve{A} = [A \ \tilde{A}] \in M_n(\mathbb{R})$, $\tilde{A}$ is arbitrary. Then $\exists\, Q, \breve{R} \in M_n(\mathbb{R})$ such that $\breve{A} = Q\breve{R}$, $\breve{R}$ is upper triangular.

Write
$$\breve{R} = [R \ T], \quad R \in M_{n,m}(\mathbb{R})$$

Then $A = QR$, $\breve{R}$ upper triangular.

$$\Rightarrow R = \begin{bmatrix} \hat{R} \\ 0 \end{bmatrix}, \quad \hat{R} \text{ is upper triangular for some}$$

■

<u>Theorem 3.3.6</u>

Let $A \in M_{n,m}(\mathbb{R})$, $b \in \mathbb{R}^n$, $n > m$ and $\operatorname{rank} A = m$. Then the LSP $Ax = b$ has a unique solution that can be found by solving the system

$$\hat{R}x = \hat{c},$$

where
$$c := Q^T b = \begin{bmatrix} \hat{c} \\ d \end{bmatrix} \in \mathbb{R}^n$$

$$\hat{c} \in \mathbb{R}^m$$

$$R = \begin{bmatrix} \hat{R} \\ 0 \end{bmatrix} \quad \text{(like in Thm 3.3.3)}$$

$Q \in M_n(\mathbb{R})$ is the orthogonal matrix such that $A = QR$.

<u>Proof</u> Consider $A = QR$ as in Thm. 3.3

Then $Ax = b$ is equivalent to

$$Q^T A x = Q^T b \qquad (Q^T = Q^{-1})$$

i.e. $Rx = c \quad (c := Q^T b)$

Write $c = \begin{bmatrix} \hat{c} \\ d \end{bmatrix}$, $\hat{c} \in \mathbb{R}^m$

& let $s := c - Rx$

Then
$$s = \begin{bmatrix} \hat{c} \\ d \end{bmatrix} - \begin{bmatrix} \hat{R} \\ 0 \end{bmatrix} x = \begin{bmatrix} \hat{c} - \hat{R}x \\ d \end{bmatrix}.$$

So
$$\|s\|_2^2 = \sum_{i=1}^{n} |s_i|^2 = \|\hat{c} - \hat{R}x\|_2^2 + \|d\|_2^2$$

Since $\|d\|_2$ does not depend on x,

$\|s\|_2$ is minimized if $\|\hat{c} - \hat{R}x\|_2$ is minimal

However, $\hat{R}$ is nonsingular. ($m \times m$, rank $\hat{R} = m$),

so $\hat{c} = \hat{R}x$ has a unique solution which therefore must be the minimizer of $\|\hat{c} - \hat{R}x\|_2$

(uniqueness follows because $\hat{c} = \hat{R}x$ has unique solution $x = \hat{R}^{-1}c$) ∎

Example

"Solve" the system $\begin{cases} x = 6 \\ 2x = 10 \end{cases}$

LSP

$$\underbrace{\begin{bmatrix} 1 \\ 2 \end{bmatrix}}_{\substack{A \\ 2\times 1}} \underbrace{[x]}_{1\times 1} = \underbrace{\begin{bmatrix} 6 \\ 10 \end{bmatrix}}_{\substack{b \\ 2\times 1}}$$

Find "x" that minimizes $\left\| \begin{bmatrix} 6 \\ 10 \end{bmatrix} - \begin{bmatrix} 1 \\ 2 \end{bmatrix} x \right\|_2$

Find QR of $A = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

Choose Q^T so that

$$Q^T A = \begin{bmatrix} * \\ 0 \end{bmatrix}$$

(see our reflector / rotation choices)

$$Q = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$$

action

$$Q^T A = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 6 \\ 10 \end{bmatrix}$$

$$= \begin{bmatrix} \sqrt{5} \\ 0 \end{bmatrix} = R$$

$$c = Q^T b = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 6 \\ 10 \end{bmatrix} = \frac{1}{\sqrt{5}} \begin{bmatrix} 26 \\ 2 \end{bmatrix}$$

By Thm. 3.3.6, we need to solve

$$\hat{R}x = \hat{c}$$

when $\hat{R} = \sqrt{5}$, $\hat{c} = \frac{1}{\sqrt{5}} 26$

i.e.

$$x = \frac{1}{\sqrt{5}} \frac{1}{\sqrt{5}} 26 = \frac{26}{5}$$

$$\boxed{x = \frac{26}{5}}$$

This x satisfies <u>none</u> of the two equations. But somehow, it minimizes the norm of the residual, so it's the LSS.

§ 3.4 The Gram-Schmidt Process

(Connection to QR factorization)

<u>Terminology</u>: $\{q_1, q_2, \ldots, q_k\} \in \mathbb{R}^n$ is called <u>orthonormal</u> if

$$q_i^T q_j = \langle q_i, q_j \rangle = \begin{cases} 0 & \text{if } i \neq j \\ 1 & \text{if } i = j \end{cases}$$

Given any linearly independent set

$$\{w_1, w_2, \dots, w_k\} \subseteq \mathbb{R}^n$$

we can $\boxed{\text{always find}}$ an orthonormal set

$$\{v_1, v_2, \dots, v_k\} \subseteq \mathbb{R}^n$$

such that

$$\text{span}\{v_1, v_2, \dots, v_m\} = \text{span}\{w_1, w_2, \dots, w_m\}$$

for all $m = 1, 2, \dots, k$.

The method/way to do it is Gram-Schmidt orthonormalization process.

$\downarrow$

<u>proof is home reading.</u>

10/9/24

- A 3rd homework set is coming up
- Take home midterm : October 30th

Food for thought based on our last example:

Recall : $\begin{cases} x = 6 \\ 2x = 10 \end{cases}$ (i)

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix} [x] = \begin{bmatrix} 6 \\ 10 \end{bmatrix}$$

$$\frac{1}{\sqrt{5}} \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} \sqrt{5} \\ 0 \end{bmatrix} [x] = \begin{bmatrix} 6 \\ 10 \end{bmatrix} \rightsquigarrow \underset{\hat{c}}{\underset{\shortparallel}{x}} = \frac{26}{5}$$

Alter the system :

$\begin{cases} x = 6 \\ x = 5 \end{cases}$ (ii)

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix} [x] = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} \sqrt{2} \\ 0 \end{bmatrix} = \begin{bmatrix} 6 \\ 5 \end{bmatrix}$$

Now $x = \frac{11}{\sqrt{2}} \neq \frac{26}{5}$

<u>Question</u> : Think about the qualative difference between (i) & (ii). Compute the residual error in both cases:

$\|b - Ax\|_2$ for the solution x

Details on Gram-Schmidt

<u>Theorem</u> Let $S = \{w_1, w_2, \ldots, w_m\} \subseteq \mathbb{R}^n$ be a linearly independent set. Define

$$S' = \{v_1, v_2, \ldots, v_m\}$$

where

$$v_1 = w_1$$

$$v_2 = w_2 - \frac{\langle w_2, v_1 \rangle}{\langle v_1, v_1 \rangle} v_1$$

$$\vdots$$

$$v_k = w_k - \sum_{j=1}^{k-1} \frac{\langle w_k, v_j \rangle}{\langle v_j, v_j \rangle} v_j$$

$$k = 1, 2, \ldots, m$$

Then S' is an orthogonal set (of nonzero vectors) and

$$\text{span}\{w_1, \ldots, w_k\} = \text{span}\{v_1, v_2, \ldots, v_k\}$$

for all $k = 1, 2, \ldots, m$

(Constructing S' is the "Gram-Schmidt orthogonalization process")

<u>Proof</u> Look at our book / 420 / 511

<u>Note</u> After applying G-S to produce S', we can "normalize":

$$\underbrace{\left\{ \frac{v_1}{\|v_1\|_2}, \frac{v_2}{\|v_2\|_2}, \dots, \frac{v_m}{\|v_m\|_2} \right\}}_{\text{orthonormal basis for } \operatorname{span}\{v_1, v_2, \dots, v_m\}}$$

<u>Example</u> $w_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$, $w_2 = \begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \end{bmatrix}$, $w_3 = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix}$

G-S.

$$v_1 = w_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$v_2 = w_2 - \frac{\langle w_2, v_1 \rangle}{\langle v_1, v_1 \rangle} v_1 = \begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \end{bmatrix} - \frac{0}{\langle v_1, v_1 \rangle} v_1$$

$$= \begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \end{bmatrix} = w_2$$

$$v_3 = w_3 - \frac{\langle w_3, v_1 \rangle}{\langle v_1, v_1 \rangle} v_1 - \frac{\langle w_2, v_2 \rangle}{\langle v_2, v_2 \rangle} v_2$$

$$= \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix} - \frac{2}{4} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} - \frac{-1}{2} \begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 0 \\ 1/2 \\ -1/2 \end{bmatrix}$$

Normalize:

$$\left\{ \frac{1}{\sqrt{4}} v_1, \ \frac{1}{\sqrt{2}} v_2, \ \sqrt{2} v_3 \right\}$$

orthonormal basis for span$\{w_1, w_2, w_3\}$

$$\left(\|v_3\|_2 = \sqrt{\frac{1}{4} + \frac{1}{4}} = \sqrt{1/2} = \frac{1}{\sqrt{2}} \right.$$

$$\left. \text{so} \quad \frac{1}{\|v_3\|_2} = \sqrt{2} \right)$$

Connection between G-S & QR

Recall in G-S:

$$v_k = w_k - \sum_{j=1}^{k-1} r_{jk} v_j$$

$$r_{jk} = \frac{\langle w_k, v_j \rangle}{\langle v_j, v_j \rangle}$$

So

$$w_1 = v_1 = r_{11} v_1$$

$$w_2 = r_{22} v_2 + r_{12} v_1$$

$$w_3 = r_{33} v_3 + r_{13} v_1 + r_{23} v_2$$

$$\vdots$$

$\rightsquigarrow$

That is

$$A = [w_1 | w_2 | \cdots | w_m]$$

$$= \underbrace{[v_1 | v_2 | \cdots | v_m]}_{\substack{n \times m \\ \text{with orthogonal} \\ \text{columns}}} \underbrace{\begin{bmatrix} r_{11} & r_{12} & \cdots & r_{1m} \\ 0 & r_{22} & r_{23} \cdots & r_{2m} \\ \cdots & & & \\ 0 & 0 \cdots & 0 & r_{mm} \end{bmatrix}}_{\substack{\text{upper} \\ \text{triangular}}}$$

$$= Q R$$

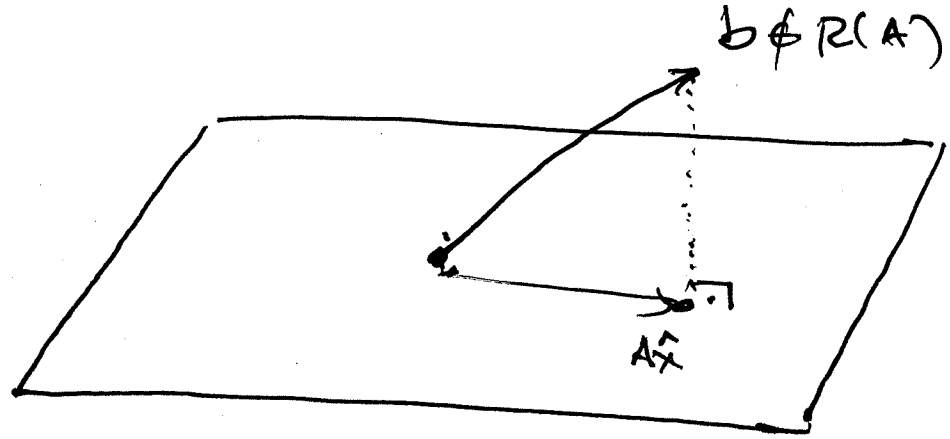
So we produced a QR-factorization of a rectangular matrix A (if $n = m$, it coincides with our past discussion)

<u>Comment</u> So now we have 3 ways of computing QR-factorization.

§ 3.5 Geometric approach to LSP

Solve $Ax = b$

$Ax = b$ is inconsistent $b \notin R(A)$



$R(A)$ ↓ column space ≈ range of A

LSP : Find $\hat{x}$ such that

$$\|A\hat{x} - b\|_2 \text{ is minimal}$$

It turns out : $A\hat{x}$ = orthogonal projection of b onto $R(A)$

Solve this system : $\hat{x}$ = LSS.

So we need to find orthogonal projections!

↓

next endeavor!

<u>Definition</u> Let $S \subseteq \mathbb{R}^n$. The <u>orthogonal complement</u>, $S^\perp$, of S is

$$S^\perp = \{ x \in \mathbb{R}^n : \langle x, y \rangle = 0, \forall y \in S \}$$

<u>Facts</u>

(1) $S^\perp$ is a subspace of $\mathbb{R}^n$

(2) $\{0\}^\perp = \mathbb{R}^n$

(3) If W is a subspace of $\mathbb{R}^n$, then $W \cap W^\perp = \{0\}$

(4) $(S^\perp)^\perp = \operatorname{span}(S)$

because $\mathbb{R}^n$ is finite-dimensional

In infinite dimensions:

$$(S^\perp)^\perp = \overline{\operatorname{span}(S)} \quad \left(\begin{array}{c}\text{topological}\\ \text{closure}\end{array}\right)$$

<u>Next</u> Orthogonal Decomposition Theorem.

10/11/24

§ 3.5 (continued)

<u>Theorem</u> (Orthogonal Decomposition Thm – ODT)

Let $S \subseteq \mathbb{R}^n$ be a subspace.
Then $\forall x \in \mathbb{R}^n$, $\exists$ unique $s \in S$, $\hat{s} \in S^\perp$ such that $x = s + \hat{s}$. Moreover,

$$s = \sum_{i=1}^{k} \langle x, v_i \rangle v_i$$

where $\{v_1, v_2, \ldots, v_k\}$ is an orthonormal basis for S, and $\hat{s} = x - s$

(s = "orthogonal projection of x onto S)

<u>Proof</u> Let $\{v_1, v_2, \ldots, v_k\}$ be an orthonormal basis of S. Then for any given $x \in \mathbb{R}^n$,

$$s := \sum_{i=1}^{k} \langle x, v_i \rangle v_i \in S$$

Let $\hat{s} = x - s$. We'll show $\hat{s} \in S^\perp$. Indeed,

$$\langle \hat{s}, v_j \rangle = \langle x, v_j \rangle - \langle s, v_j \rangle$$

$$= \langle x, v_j \rangle - \langle \sum_{i=1}^{n} \langle x, v_i \rangle v_i, v_j \rangle$$

$$= \langle x, v_j \rangle - \sum_{i=1}^{k} \langle x, v_i \rangle \langle v_i, v_j \rangle$$

$$\overset{!}{=} \langle x, v_j \rangle - \langle x, v_j \rangle = 0$$

That $\hat{s} \perp v_j$, $j = 1, 2, \ldots, n$; thus $\hat{s} \in S^\perp$

So we've shown: $s \in S$, $\hat{s} \in S^\perp$, $x = s + \hat{s}$.

To prove uniqueness of $s, \hat{s}$: Assume

$$x = s + \hat{s} = s_1 + \hat{s}_1$$

where $s, s_1 \in S$, $\hat{s}, \hat{s}_1 \in S^\perp$

Then

$$\underbrace{s - s_1}_{\in S} = \underbrace{\hat{s}_1 - \hat{s}}_{\in S^\perp} \quad \& \quad S \cap S^\perp = \{0\}$$

Thus $s - s_1 = \hat{s}_1 - \hat{s} = 0$, i.e. $s_1 = s$, $\hat{s}_1 = \hat{s}$ ∎

<u>Example</u> Let $S = \text{span}\left\{ \underbrace{\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}}_{\text{basis for } S} \right\}$

Let $x = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$. Write $x = s + \hat{s}$, $s \in S$, $\hat{s} \in S^\perp$.

To use ODT, we need an <u>orthonormal</u> basis for S. Find one via G-S:

<u>G-S</u> gives

$$v_1 = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\1\\0\end{bmatrix}, \quad v_2 = -\frac{1}{\sqrt{6}}\begin{bmatrix}1\\-1\\2\end{bmatrix}$$

$$S = \text{span}\{v_1, v_2\} \ \& \ v_1 \perp v_2$$
$$\& \ \|v_1\|_2 = \|v_2\|_2 = 1.$$

Apply ODT $\left(x = \begin{bmatrix}0\\1\\1\end{bmatrix}\right)$

$$s = \langle x, v_1\rangle v_1 + \langle x, v_2\rangle v_2$$

$$= \frac{1}{\sqrt{2}}\begin{bmatrix}1/\sqrt{2}\\1/\sqrt{2}\\0\end{bmatrix} + \frac{1}{\sqrt{6}}\begin{bmatrix}1/\sqrt{6}\\-1/\sqrt{6}\\2/\sqrt{6}\end{bmatrix}$$

$$= \cdots = \frac{1}{3}\begin{bmatrix}2\\1\\1\end{bmatrix} \in S$$

$$\hat{s} = x - s = \cdots = \frac{1}{3}\begin{bmatrix}-2\\2\\2\end{bmatrix} \in S^{\perp}$$

$$x = s + \hat{s} \ \checkmark$$

Do the same for $y = \begin{bmatrix}2\\1\\1\end{bmatrix} = t + \hat{t}$,
$t \in S$, $\hat{t} \in S^{\perp}$.

<u>Guess $t, \hat{t}$</u> $\left(S = \text{span}\left\{\begin{bmatrix}1\\1\\0\end{bmatrix}, \begin{bmatrix}1\\0\\1\end{bmatrix}\right\}\right)$

↳ Observe $y = \begin{bmatrix}1\\1\\0\end{bmatrix} + \begin{bmatrix}1\\0\\1\end{bmatrix} \in S$

So $t = \begin{bmatrix}2\\1\\1\end{bmatrix} = y$, $\hat{t} = y - t = 0$.
(confirm via ODT)

<u>Corollary</u> (of ODT $\rightsquigarrow$ applies to LSP)

Let S be a subspace of $\mathbb{R}^n$. Let $x \in \mathbb{R}^n$ be decomposed as

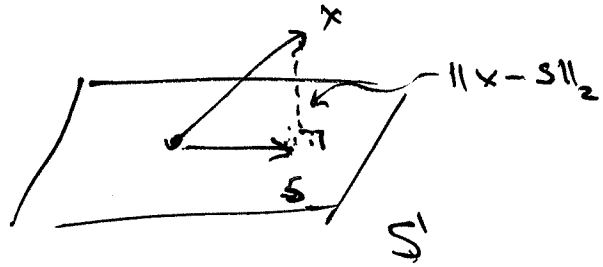
$$x = s + \hat{s}, \quad s \in S, \quad \hat{s} \in S^{\perp}.$$

Then $\forall y \in S$,

$$\|x - y\|_2 \geq \|x - s\|_2 .$$

Equality occurs iff $y = s$.

(Interpretation: s = orth. proj. of x onto S
= vector in S closest to x)

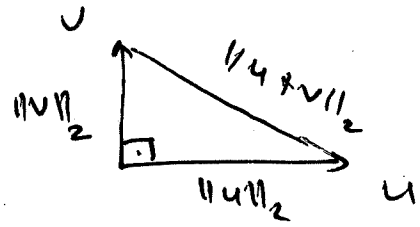


<u>Proof</u>

Recall Pythagorean Theorem in ips

$u, v \in \mathbb{R}^n$: $u \perp v \iff \|u+v\|_2^2 = \|u\|_2^2 + \|v\|_2^2$

prove!



So to prove the corollary:

$$x = s + \hat{s}, \quad s \perp \hat{s}$$

Then for any $y \in S$,

$$y - s \perp \hat{s} \qquad (y \in S)$$

Then
$$\begin{aligned}
\|x-y\|_2^2 &= \|s+\hat{s}-y\|_2^2 \\
&= \|(s-y)+\hat{s}\|_2^2 \\
&\overset{\text{Pyth.}}{=} \underbrace{\|s-y\|_2^2}_{\geq 0} + \|\hat{s}\|_2^2 \\
&\geq \|\hat{s}\|_2^2 = \|x-s\|_2^2 .
\end{aligned}$$

$$\Rightarrow \|x-y\|_2 \geq \|x-s\|_2 \checkmark$$

Equality holds iff $\|s-y\|_2 = 0$, i.e. $s-y=0$, i.e. $s=y$ ∎

LSP

Let's use the corollary to solve LSP:

$$Ax = b$$

Minimize $\|b - Ax\|_2$ over $x \in \mathbb{R}^n$

That is, find

$$\min_{s \in R(A)} \|b - s\|_2$$

where $R(A) = \{Ax : x \in \mathbb{R}^n\}$

By previous corollary, the solution $\hat{x}$ to this minimization problem satisfies

$A\hat{x} = s =$ orthog. projection of b onto $R(A)$

So to solve LSP $Ax = b$:

1. Find $b = s + \hat{s}$, $s \in R(A)$, $\hat{s} \in R(A)^{\perp}$
2. Solve $A\hat{x} = s$ for $\hat{x}$